Internet of Things Course (420-521-VA)

Final Project Phase 4

Due Date:

Monday, December 1st , 2025

Team Members (Group 5)

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Overview

Phase 4 focused on finalizing the entire Smart Store system by integrating all features from Phases 2 and 3 and implementing the full Report Generation Module. The goal was to ensure a complete end-to-end workflow for administrators and customers, including inventory reports, sales analytics, and customer purchase history. All modules were polished and prepared for the final demonstration.

System Design

During Phase 4, the system architecture was expanded to support dynamic reporting for both admin and customer interfaces.

The backend now queries real-time inventory, sales, and customer activity data, while the frontend displays results using tables and visual elements.

Customers can now view receipt history, spending totals, and search for past purchases. Administrators can view inventory levels, sales performance, and customer activity within selected date ranges.

Data Flow

• Purchases update inventory levels instantly and are logged for reporting.
• Admin reports fetch data filtered by date, product, or customer activity.
• Customer purchases are saved with timestamps and item details for receipt history.
• Search tools allow customers to find how many times they bought a specific item and at what price.
• All reporting pages read directly from Supabase to ensure accuracy and consistency across the system.

Implementation

• Added reporting routes and templates for admin and customer dashboards.
• Implemented Inventory Report with real-time stock levels and optional low-stock alerts.
• Implemented Sales Report with sold-item counts, top/least sellers, and total revenue for any date range.

• Implemented Customer Activity Report with metrics on daily or range-based customer purchases.
• Added customer receipt history, total spending calculation, and item search functionality.
• Improved UI consistency, added optional theme support, and polished bilingual pages.
• Ensured all Phase 2 and 3 features integrate cleanly with the new reporting tools.

- Repository: https://github.com/Crisguts/IoT-Semester-Project-

Task Distribution

Team Member	Task
Cristian	Merged all branches and integrated Parth's inventory/activity reporting and Adlane's sales report into the final system. Implemented light/dark mode, updated all English/French translations, created the entire customer-side reporting module, and added enhancements across all report features.
Trevor	Updated UI/UX issues across all phases, and his design style was applied throughout Phase 4.
Parth	Designed and implemented the full administrator Inventory Report and Activity Report modules with advanced filtering, accurate real-time stock tracking, and seamless database synchronization. Developed the backend logic powering product-level availability, customer activity metrics, and date-range analytics. Introduced report to PDF tool to be used by the other report modules.
Adlane	Implemented the entire Sales Report feature and page, including date selection and filters. Implemented total sales value for the selected time period, created the sold-items-by-product table with pagination support, and added tables showing the top 3 and bottom 3 selling items.